

proprietary interpreters. It only does what it is instructed to do in a way one expects it to be done. Nagios lets the user intelligently schedule monitoring programs in any language. These monitoring programs are called plugins, which report the monitoring status back to Nagios. It has lots of hooks that make it easy to get data in and out; so it can provide real-time data to graphing programs such as RRDTool and MRTG, and can easily work along with other monitoring systems, either by feeding them or by being fed by them.

Nagios Core and Nagios XI

Nagios Core is open source whereas Nagios XI is a commercial, enterprise version of Nagios. Unlike Nagios Core, Nagios XI is completely Web based. It is also wizard driven. The dashboards are more robust and personalised. There are plenty of additional features in the enterprise version such as an advanced UI, maintenance tools, integration with third party ticketing and much more.

How does Nagios work?

Nagios was developed for Linux but is now capable of monitoring UNIX based operating systems. It has three components — the daemon, the Web interface, and the plugins. Nagios has four basic object types, which are listed below.

Commands: These are used to interact with the plugins, and to control the event handlers, notifications and checks.

Contact and contact groups: These define the individuals to be contacted in case of an event.

Host and host groups: These are used to specify services and hosts in a particular network.

Notifications: These define what content needs to be sent to contacts and contact groups when there are drifts.

Services: These specify network services such as HTTP/80, SSH, FTP, etc, and host services such as clock, DNS entries, etc.

Host and service check definitions instruct Nagios which plugins to call to obtain the status of a host or service. The definitions in the plugins are checked against that particular host/service. If it is seen going beyond the threshold, Nagios notifies the contacts/contact groups. Nagios returns any one of the following four status codes for any event:

- 0-OK
- 1-Warning
- 2-Critical
- 3-Unknown

Nagios can be configured to do either passive checks or active checks. In active checks, the host that runs Nagios 'polls' devices or services for status information every so often, whereas passive checks are initiated and performed by external applications/processes. The results are then sent to the monitoring host (where Nagios runs) for processing. The major difference between active and passive checks is that active checks are initiated and performed by Nagios, while passive checks are performed by external applications. Passive

checks can be used to monitor asynchronous services and to monitor the services that are located behind a firewall. Active checks can be used for on-demand checks and regular interval checks. The results can be monitored via the Web GUI, which is CGI based.

Features

The features of Nagios are listed below:

- Centralised view of the entire monitored IT infrastructure
- Automatic restart of failed applications is granted by its event handlers
- Multi-user access
- Selective access allows clients to view only the infrastructure components that affect them
- Monitoring of network services like SMTP, POP3, HTTP, NNTP, PING, etc
- Monitoring of host resources such as processor load, disk usage, etc
- Parallelised service checks
- Ability to define network host hierarchy using 'parent' hosts, allowing detection of and distinction between hosts that are down and those that are unreachable
- Optional Web interface for viewing current network status, notifications and problem history, log files, etc
- Powerful script APIs allow easy monitoring of in-house and custom applications, services, and systems
- Availability reports ensure SLAs are being met
- Failover capabilities ensure non-stop monitoring of critical IT infrastructure components

Installation

Let's install Nagios on Ubuntu 16.x (or above) where SELinux is not in an enforcing state.

Prerequisite: Graphics library to display the monitoring on the Web front-end (Apache). Use the following commands to download it:

```
$sudo apt-get update
$sudo apt-get install -y autoconf gcc libc6 make wget unzip
apache2 php libapache2-mod-php7.0 libgd2-xpm-dev
```

Download Nagios from the source, as follows:

```
$cd /tmp
$wget -O nagioscore.tar.gz https://github.com/
NagiosEnterprises/nagioscore/archive/nagios-4.3.4.tar.gz
$tar xzf nagioscore.tar.gz
```

Compile the source using the following code:

```
$cd /tmp/nagioscore-nagios-4.3.4/
$sudo ./configure --with-httpd-conf=/etc/apache2/sites-
enabled
$sudo make all
```